Ajudiya Sneh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.02
Nand Vidya Niketan, Jamnagar, Gujarat	Class XII (CBSE)	2023	95.00%
Nand Vidya Niketan, Jamnagar, Gujarat	Class X (CBSE)	2021	96.80%

WORK EXPERIENCE

Full Stack Developer Intern | NRI PelliGola

- Built an automated **weekly digest** email system with **cron scheduling**, reaching **~2K** users with activity wrap-ups.
- Developed paid-gated **profile-analytics chart**, used by users to track a total of **80K+ visit events** across all profiles.
- Built the **block-profile feature** that improved user safety, preferences and control, generating **600+** block records.
- Worked on **18+** other major tasks including new features, functionality updates and bug fixing across the whole platform.

Web Developer Intern | CMARIX Infotech

- Designed role-driven e-commerce modules with **Admin, Vendor, Customer** workflows and dashboards.
- Secured the platform via JWT **authentication**, middleware-based **authorization**, and centralized **error handling** logic.
- Enabled catalog, cart, checkout, and subscription billing via **secure Stripe workflows** backed by MongoDB Atlas.
- Built real-time 1-on-1 and broadcast chats for issues handling using **websockets** with room-based architecture.

PROJECTS

Adaptive In-Process Cache Github

- Built a thread-safe C++ cache library with byte-bounded intrusive LRU, avoiding list overhead for **O(1) ops**.
- Designed **ShardedCache** with per-shard shared_mutex and exclusive-lock, since node promotion mutates pointers.
- Added **TinyLFU** admission via lock-free **CountMinSketch**, resisting scan pollution and lifting **hit rate** up to **4.05%**.
- Benchmarked at **16** threads: **9.6-10.5M ops/sec** with zero TSan races, diagnosing hot-shard lock contention plateau.

Async TCP Server with MLFQ Task Scheduler Github

- Built an async **epoll TCP reactor** paired with a 3-level **MLFQ scheduler** using std::jthread worker thread pools.
- Designed **quantum-based demotion** at 5ms, 20ms, and 100ms with a periodic **priority booster** to prevent starvation.
- Benchmarked **16** concurrent clients, **2458 requests, 0 errors**, keeping light-task P50 latency at **3.6ms** in Queue 0.
- Validated anti-head-of-line blocking via **1097 heavy-task demotions**, sustaining **401 req/sec** aggregate throughput.

Dopamine Receptor Activity Prediction GitHub

- Built an **sklearn regression pipeline** predicting dopamine D1-D5 receptor pEC50 potency from molecular descriptors.
- Trained **14+ regressors** (Ridge, SVR, RF, AdaBoost, GB, XGBoost), **tuned** via RandomSearchCV and GridSearchCV.
- **Resolved overfitting** via early stopping and feature-importance analysis, achieving **96%** cross-validated R2 score.

Financial Order Book Github

- Built a non-recursive iterative **segment tree** using **bitwise** index traversal for highly optimized tick updates.
- Designed a monoid node combining min price, max price, volume, and dollar volume for **O(log N)** range merges.
- Implemented **VWAP** as a derived monoid field with zero-division-safe identity and multi-trade tick accumulation.
- Sustained **10.58M** point updates/sec and **150ns** range-query median across **10-20K ticks**, confirming logarithmic scaling.

SKILLS

Programming Languages: C, C++, Python, JavaScript/Typescript

Web Technologies: Node.js, Express, React, REST APIs, HTML, CSS

Databases: MySQL, MongoDB, PostgreSQL

Version control and Platforms: Git, GitHub, VS Code, Docker

Core Competences: DSA, OS, CN, OOPs

RELEVANT COURSES

Computer Science Electives: Introduction to Programming, Introduction to Artificial Intelligence

Mathematics Electives: Calculus I and II, Elementary Linear algebra, Differential Equations, Introduction to Probability

Other Electives: Numerical and Computational Lab, ML in Chemistry, Communication Skills, Organisational structure

POSITIONS OF RESPONSIBILITY

Core | Blockspace (The Blockchain and Crypto Club)

Volunteered in technical sessions on blockchain basics and smart contracts and Web3 talks.

Class Representative

Coordinated between academic secretary, faculty and students on academic and administrative matters.

EXTRACURRICULAR

- Participated and qualified to round 2 in all India Adobe University Hackathon 2026.
- Represented school football team at **U-17 Subroto Cup State Tournament**, showcasing team leadership and discipline.
- Secured **2nd place** in football at IIT Hyderabad's **Fresher's Tournament**, contributing as a starting player.
- Active **NSS Volunteer (2023-27)**, involved in cleanliness drives, tree plantations, and health awareness campaigns.